

CS & DA



Probability and Statistics

Statistics II

DPP- 01

Discussion Notes

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#Q. Consider two random variables X and Y with joint PMF given in Table.

$X \backslash Y$	Y=2	Y=4	Y=5
X=1	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$
X=2	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$
X=3	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$

$P(Y)$

$\frac{6}{12}$

$\frac{6}{24}$

$\frac{6}{24}$

$P(X)$

$= \frac{4}{24}$

$= \frac{9}{24}$

$= \frac{11}{24}$

1

$X :$

1	2	3
$\frac{1}{6}$	$\frac{3}{8}$	$\frac{11}{24}$

$\sum p_i = 1$

$Y :$

2	4	5
$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

$\sum p_i = 1$

1) Find $P(X \leq 2, Y \leq 4) = P[X=1 \text{ or } 2 \cap Y=2 \text{ or } 4]$

$= P[(1,2) \text{ or } (1,4) \text{ or } (2,2) \text{ or } (2,4)]$

$= \frac{1}{12} + \frac{1}{24} + \frac{1}{6} + \frac{1}{12} = \frac{2+1+4+2}{24} = \frac{9}{24} = \frac{3}{8}$

2) Find the marginal PMFs of X and Y

$X =$

$Y =$

$P(X) =$

$P(Y) =$

3) Find $P(Y=2|X=1) = \frac{P(Y=2 \cap X=1)}{P(X=1)} = \frac{P(1,2)}{P(X=1)} = \frac{1/12}{1/6} = \frac{1}{2}$

4)

Are X and Y independent?

$$P(X=x_i \cap Y=y_j) = P(X=x_i) \cdot P(Y=y_j) \quad \forall i \neq j$$

eg $P(X=2, Y=2) = P(2,2) = \left(\frac{1}{6}\right)$

$$P(X=2) \cdot P(Y=2) = \frac{3}{8} \cdot \frac{1}{2} = \frac{3}{16} \neq \frac{1}{6} \quad \text{So Not Ind}$$

#Q. For the adjoining bivariate probability distribution of X and Y, find:

$X \backslash Y$	Y=1	Y=2	Y=3	Y=4	Y=5	Y=6	$P(X)$
X=0	0	0	$1/32$	$2/32$	$2/32$	$3/32$	$8/32$
X=1	$1/16$	$1/16$	$1/8$	$1/8$	$1/8$	$1/8$	$10/16$
X=2	$1/32$	$1/32$	$1/64$	$1/64$	0	$2/64$	$8/64$
$P(Y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	

$$\begin{aligned}
 1) P(X \leq 1, Y=2) &= P(X=0 \text{ or } 1 \cap Y=2) \\
 &= P[(0,2) \text{ or } (1,2)] \\
 &= 0 + \frac{1}{16} = \frac{1}{16}
 \end{aligned}$$

$$2) P(X \leq 1) = P(X=0 \text{ or } 1) = \frac{8}{32} + \frac{10}{16} = \frac{8}{32} + \frac{20}{32} = \frac{28}{32}$$

$$3) P(Y \leq 3) = P(Y=1 \text{ or } 2 \text{ or } 3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{6+6+11}{64} = \frac{23}{64}$$

$Y:$	1	2	3	4	5	6
$P(Y):$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$

$$\begin{aligned}
 4) P(X < 3, Y \leq 4) &= P(X = 0 \text{ or } 1 \text{ or } 2 \cap Y = 1 \text{ or } 2 \text{ or } 3 \text{ or } 4) \\
 &= P(X = 0, Y \leq 4) + P(X = 1, Y \leq 4) + P(X = 2, Y \leq 4) \\
 &= \left(0 + 0 + \frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right) = \left(\frac{9}{16}\right)
 \end{aligned}$$

#Q. For the joint probability distribution of two random variables X and Y, find:

X Y	Y = 1	Y = 2	Y = 3	Y = 4	Total
X = 1	4/36	3/36	2/36	1/36	10/36
X = 2	1/36	3/36	3/36	2/36	9/36
X = 3	5/36	1/36	1/36	1/36	8/36
X = 4	1/36	2/36	1/36	5/36	9/36
Total	11/36	9/36	7/36	9/36	1

X:	1	2	3	4
P(X):	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

Y:	1	2	3	4
P(Y):	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

1) Find the marginal distributions of X and Y

2) Conditional distribution of X given the value of $Y=1$ and that of Y given the value of $X=2$

X	:	1	2	3	4
$P(X=x/Y=1)$	:	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{1}{11}$

$$P(X=1/Y=1) = \frac{P(1,1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$

$$P(X=2/Y=1) = \frac{P(2,1)}{11/36} = \frac{1/36}{11/36} = \frac{1}{11}$$

$$P(X=3/Y=1) = \frac{P(3,1)}{11/36} = \frac{5/36}{11/36} = \frac{5}{11}$$

$$P(X=4/Y=1) = \frac{1/36}{11/36}$$

Y	:	1	2	3	4
$P(Y=y/X=2)$	:	$\frac{1}{9}$	$\frac{3}{9}$	$\frac{3}{9}$	$\frac{2}{9}$

$$P(Y=1/X=2) = \frac{P(2,1)}{P(X=2)} = \frac{1/36}{9/36} = \frac{1}{9}$$

$$P(Y=2/X=2) = \frac{P(2,2)}{P(X=2)} = \frac{3/36}{9/36} = \frac{3}{9}$$

$$P(Y=3/X=2) = \frac{P(2,3)}{P(X=2)} = \frac{3/36}{9/36} = \frac{3}{9}$$

$$P(Y=4/X=2) = \frac{P(2,4)}{P(X=2)} = \frac{2/36}{9/36} = \frac{2}{9}$$

#Q. I have a bag containing 40 blue marbles and 60 red marbles. I choose 10 marbles (without replacement) at random. Let X be the number of blue marbles and Y be the number of red marbles. Find the joint PMF of X and Y .

eg 3 Blue & 4 Red

& we want to take 3 Marbles w/o Replacement.
then prob of drawing 1 B & 2 R are?

(M-I) (B R R) or (R B R) or (R R B)

$$\left(\frac{3}{7} \times \frac{4}{6} \times \frac{3}{5}\right) + \left(\frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}\right) + \left(\frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}\right) = \left(\frac{4 \times 3 \times 3}{7 \times 6 \times 5}\right) \times 3$$

~~(M-I)~~ Req Prob = $\frac{{}^3C_1 \times {}^4C_2}{{}^7C_3} = \frac{3 \times \left(\frac{4 \times 3}{2}\right)}{\frac{7 \times 6 \times 5}{3 \times 2}} = \left(\frac{3 \times 4 \times 3}{7 \times 6 \times 5}\right) \times 3$

fav Cases = $\{(0,10), (1,9), (2,8), \dots, (10,0)\}$

$$x + y = 10$$

$$P(x=x_i, y=y_j) = \frac{{}^{40}C_{x_i} {}^{60}C_{y_j}}{{}^{100}C_{10}}, x_i + y_j = 10$$

eg 0 , otherwise

$$P(x=1, y=2) = \frac{{}^3C_1 + {}^4C_2}{{}^7C_3}$$

#Q. The joint p.d.f of two random variables X and Y is given by:

$$f(x, y) = 9(1 + x + y)/2((1 + x)^4(1 + y)^4) \quad 0 \leq x < \infty, 0 < y < \infty$$

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} \frac{9(1+x+y)}{2(1+x)^4(1+y)^4} dy = \frac{9}{2(1+x)^4} \int_0^{\infty} \frac{(1+y+x)}{(1+y)^4} dy = \frac{9}{2(1+x)^4} \int_0^{\infty} \left[\frac{(1+y)^{-3}}{1} + x(1+y)^{-4} \right] dy$$

Find the marginal distributions of X and Y, and the conditional distribution of Y for $X=x$.

$$= \frac{9}{2(1+x)^4} \left[\frac{(1+y)^{-2}}{-2} + x \frac{(1+y)^{-5}}{-5} \right]_0^{\infty} = \frac{9}{2(1+x)^4} \left[\frac{-1}{2(1+y)^2} - \frac{x}{5(1+y)^5} \right]_0^{\infty} = \frac{-9}{2(1+x)^4} \left[0 - \left\{ \frac{1}{2} + \frac{x}{5} \right\} \right]$$

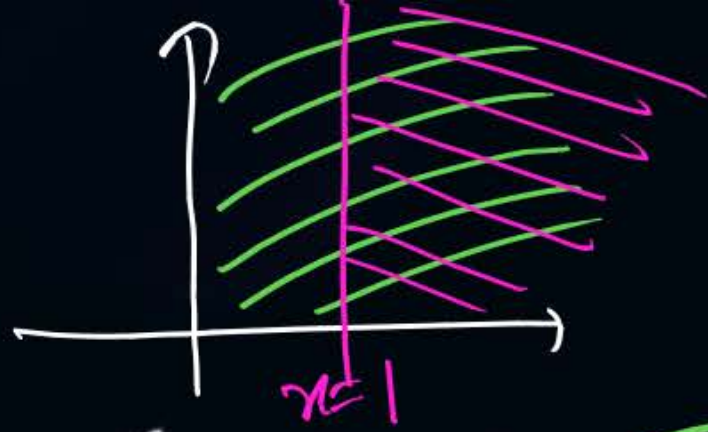
$$\text{Similarly } f(y) = \int_0^{\infty} f(x, y) dx = \dots = \frac{3}{4} \cdot \frac{(3+2y)}{(1+y)^4}, \quad 0 \leq y < \infty$$

$$\boxed{P\left[\gamma=y/x=n\right] = \frac{f(n,y)}{f(n)}} = \frac{\frac{9(1+n+y)}{2(1+n)^4(1+y)^4}}{\frac{3}{4} \cdot \frac{(3+2n)}{(1+n)^4}} = \frac{6(1+n+y)}{(1+y)^4(3+2n)}''$$

$0 \leq y < \infty$

#Q.

Given-



$$f(x, y) = \begin{cases} (e^{-(x+y)}), & 0 \leq x < \infty, 0 \leq y < \infty \end{cases}$$

Calculate the value of 1) $P(X > 1) = \int_{y=0}^{\infty} \int_{x=1}^{\infty} f(x, y) dx dy = \int_{y=0}^{\infty} \int_{x=1}^{\infty} e^{-x} \cdot e^{-y} dx dy$

$$= \int_{x=1}^{\infty} e^{-x} dx \int_{y=0}^{\infty} e^{-y} dy = \frac{1}{e} \times 1 = \left(\frac{1}{e} \right)$$

[NAT]



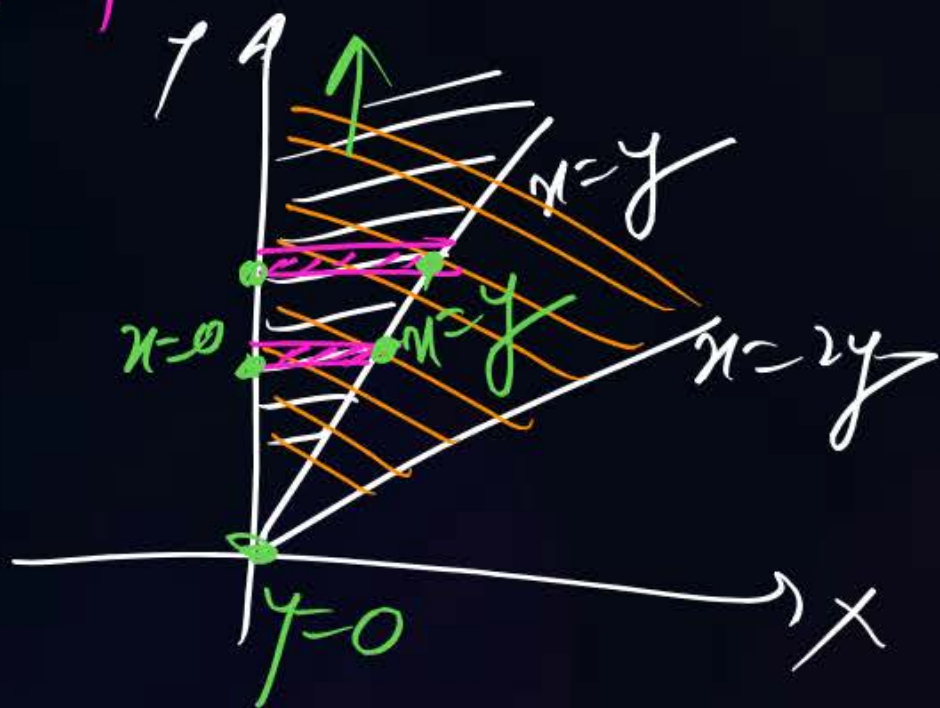
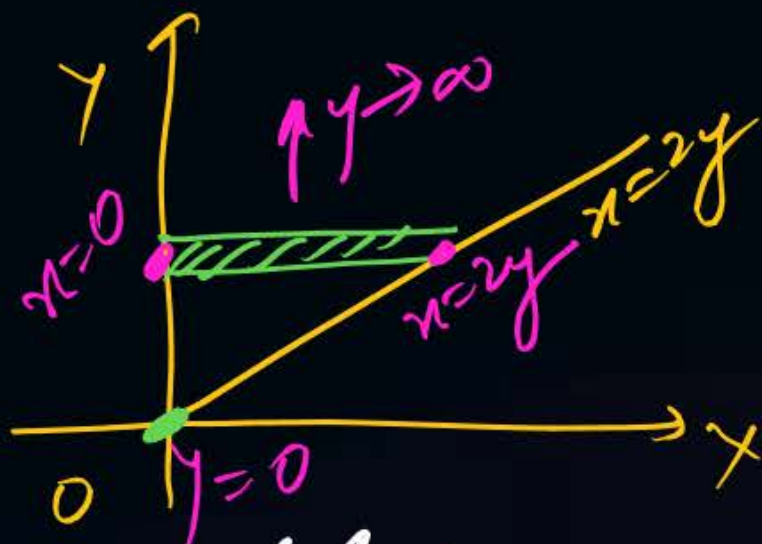
2)

$P(X < Y | X < 2Y)$

$$= \frac{P[X < Y \cap X < 2Y]}{P(X < 2Y)}$$

$$= \frac{P(X < Y)}{\int_0^\infty \int_0^{2y} f(x, y) dx dy}$$

$$= \frac{\int_0^\infty \int_0^y e^{-(x+y)} dx dy}{\int_0^\infty \int_0^{2y} e^{-(x+y)} dx dy}$$



$$= \frac{\int_0^\infty \left(\frac{e^{-x}}{-1} \right)_{x=0}^y (e^{-y} dy)}{\int_0^\infty \left(\frac{e^{-x}}{-1} \right)_{x=0}^{2y} (e^{-y} dy)}$$

$$= \frac{\int_0^\infty (1 - e^{-y}) e^{-y} dy}{\int_0^\infty (1 - e^{-2y}) e^{-y} dy}$$

$$= \frac{\int_0^\infty (e^{-y} - e^{-2y}) dy}{\int_0^\infty (e^{-y} - e^{-3y}) dy}$$

$$= \frac{1/2}{2/3} = \frac{3}{4} \text{ Ans}$$

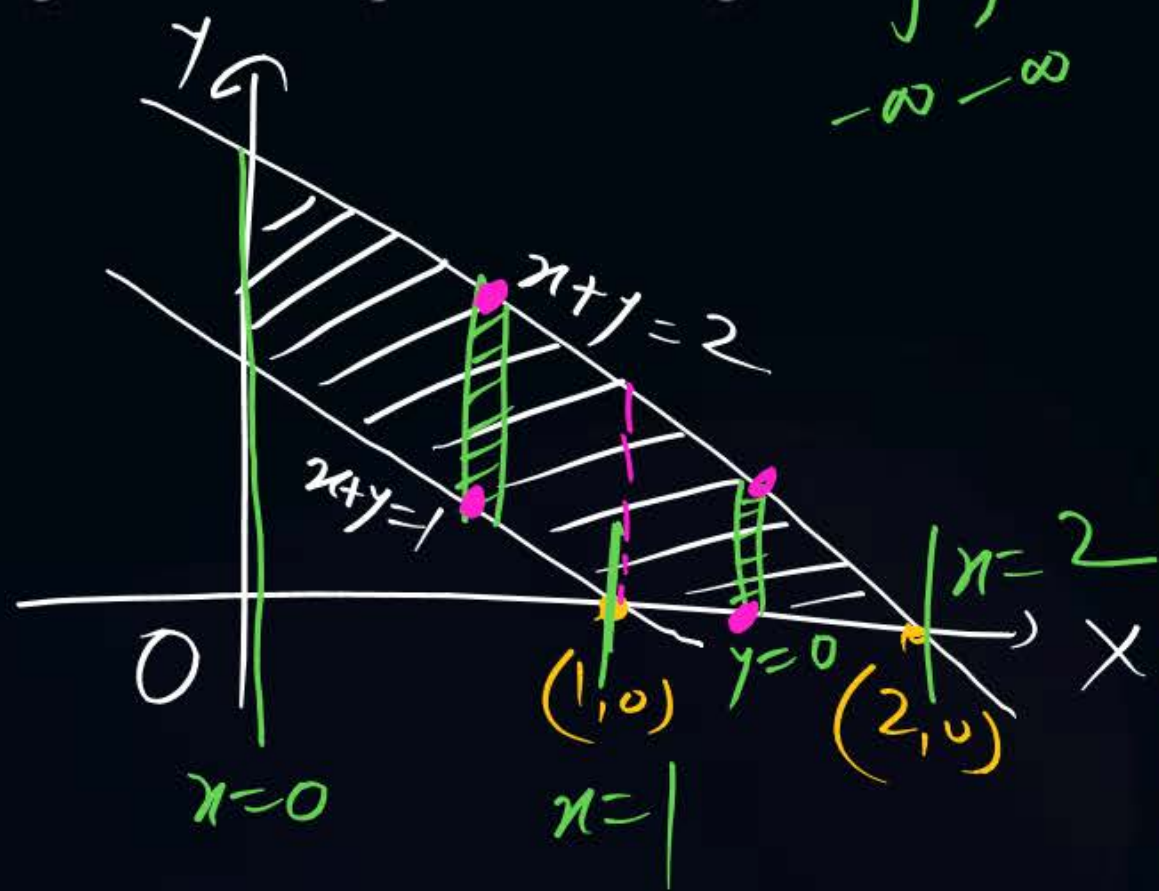
[NAT]



3)

$$P(1 < X+Y < 2)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = \int_{x=0}^1 \int_{y=1-x}^{2-x} e^{-(x+y)} dy dx + \int_{x=1}^2 \int_{y=0}^{2-x} e^{-(x+y)} dy dx$$



$$= \int_{x=0}^1 \left(\frac{e^{-y}}{-1} \right)_{y=1-x}^{2-x} (e^{-x} dx) + \int_{x=1}^2 \left(\frac{e^{-y}}{-1} \right)_0^{2-x} (e^{-x} dx)$$

$$= \int_0^1 (e^{x-1} - e^{x-2}) e^{-x} dx + \int_1^2 (1 - e^{x-2}) e^{-x} dx$$

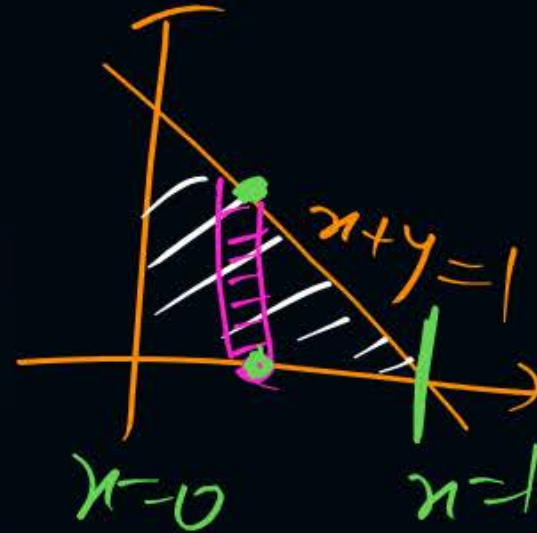
$$= \int_0^1 (e^{-1} - e^{-2}) dx + \int_1^2 (e^{-x} - e^{-2}) dx$$

$$= \frac{2}{e} - \frac{3}{e^2}$$

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cx + 1 & x, y \geq 0, x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

1) Find the constant c .



w.k. that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$

$$\int_{x=0}^1 \int_{y=0}^{y=1-x} (cx+1) dy dx = 1$$

$$\int_{x=0}^1 (1-x)(cx+1) dx = 1$$

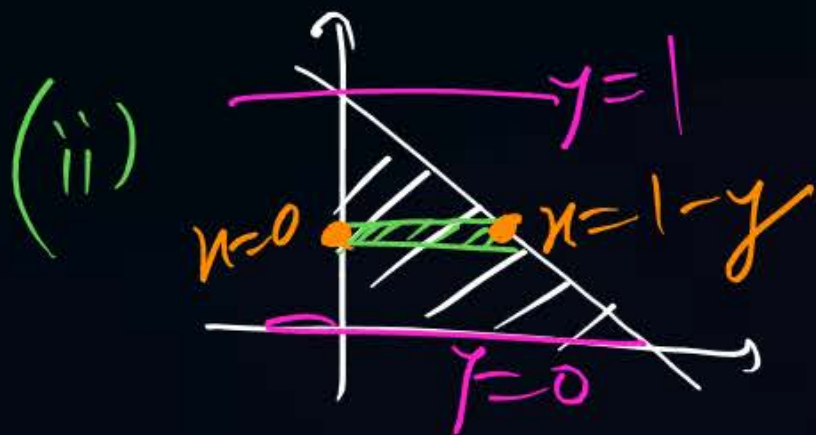
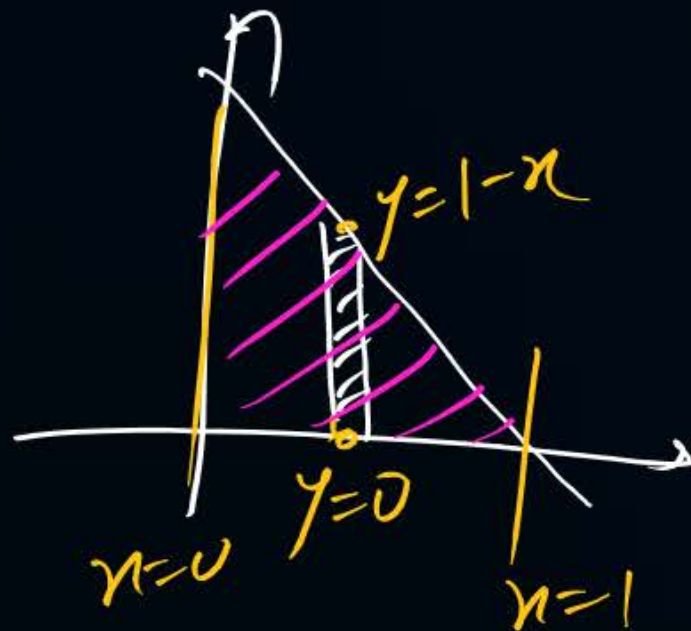
$$c=3$$

[NAT]



2) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_{y=0}^{1-x} (3x+1) dy = \boxed{(3x+1)(1-x)} \quad 0 \leq x \leq 1$$



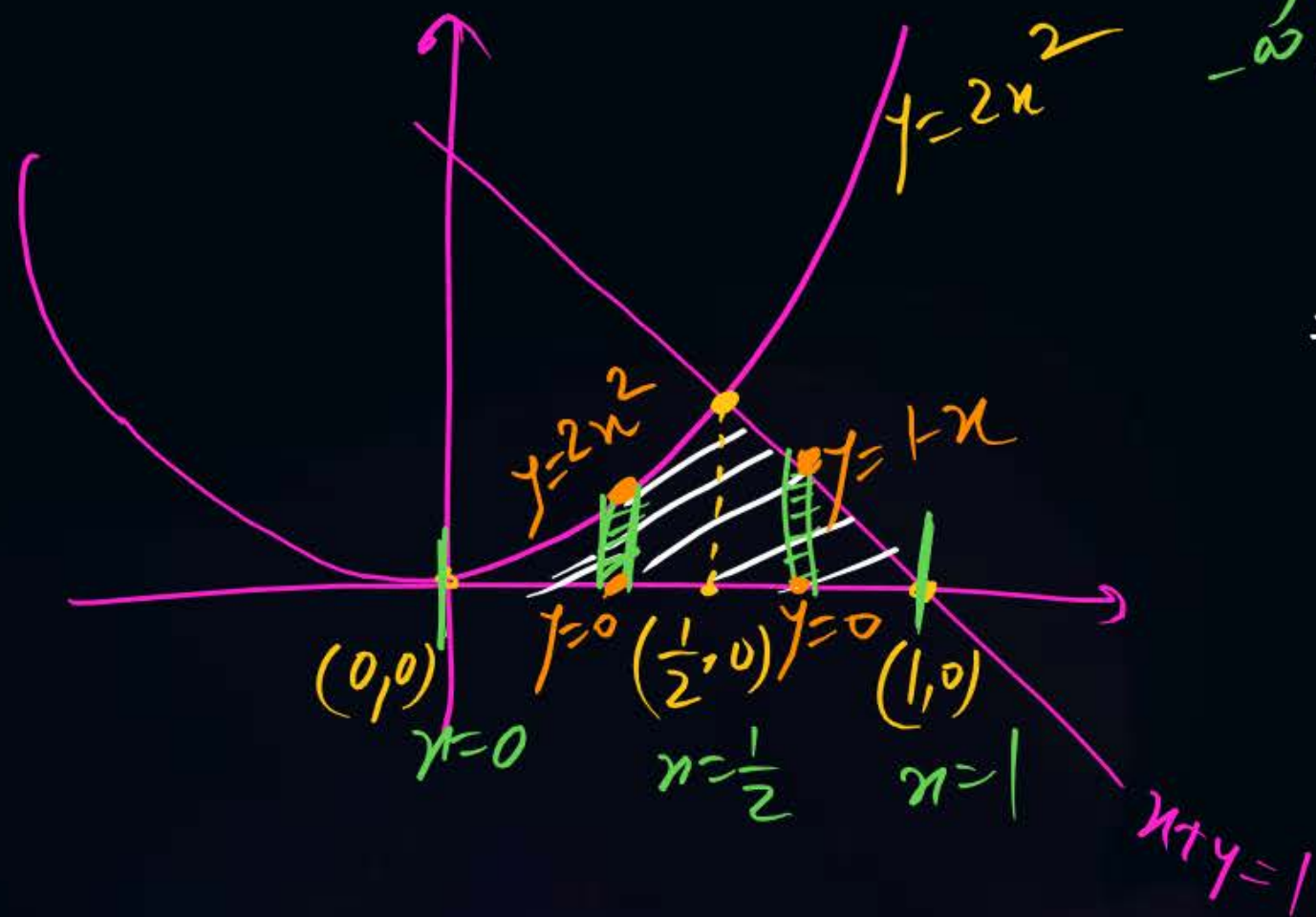
$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{x=0}^{1-y} (3x+1) dx = \left[3\left(\frac{x^2}{2}\right) + x \right]_0^{1-y} = \frac{3}{2}(1-y)^2 + (1-y) \\ = \boxed{\frac{1}{2}(1-y)(5-3y)} \\ 0 \leq y < \infty$$

[NAT]



3)

Find $P(Y < 2X^2)$.



$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (3x+1) dx dy = \int_{x=0}^{\frac{1}{2}} \int_{y=0}^{2x} (3x+1) dy dx + \int_{x=\frac{1}{2}}^1 \int_{y=0}^{1-x} (3x+1) dy dx$$

$$= \int_{x=0}^{\frac{1}{2}} (3x+1)(2x) dx + \int_{x=\frac{1}{2}}^1 (3x+1)(1-x) dx$$

$$= 53/96$$

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$f(x,y) = 6e^{-2x-3y}$
 $= (2e^{-2x})(3e^{-3y})$
 $= f(x)f(y)$

1) Are X and Y independent? **YES**

if there are **Ind**

E-Disst: $f(t) = \begin{cases} \mu e^{-\mu t}, & t \geq 0 \\ 0, & \text{else} \end{cases}$

$\Rightarrow \mu = 3 \text{ \& } E(t) = \frac{1}{\mu} = \frac{1}{3}$

Good check: $f(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^{\infty} 6e^{-2x-3y} dy$
 $= 2e^{-2x}$
 $\& f(y) = 3e^{-3y}, y \geq 0$

2) Find $E[Y|X > 2] = E(Y) = \frac{1}{3}$

w. k that In Case of Independency, Condition has No significance.

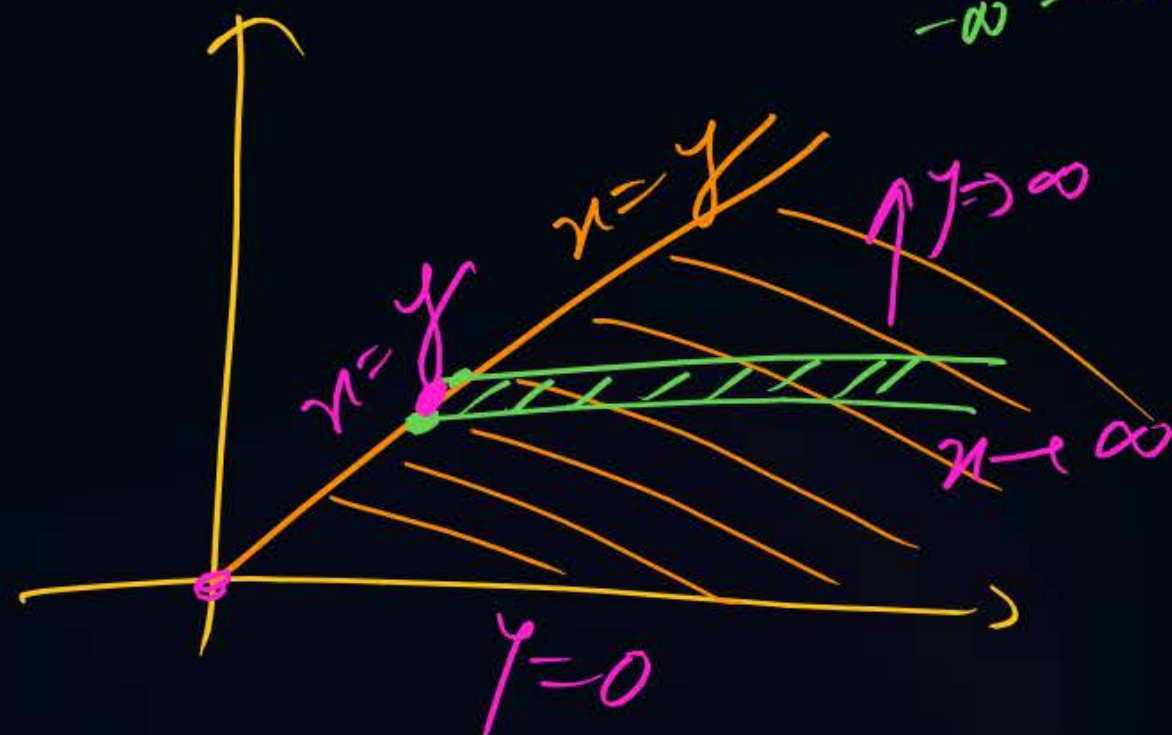
Similarly $E(X) = \frac{1}{2}$

[NAT]



3)

Find $P(X > Y)$.



$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_{y=0}^{\infty} \int_{x=y}^{\infty} (6 e^{-2x} \cdot e^{-3y}) dx dy$$

$$= \int_{y=0}^{\infty} \left(\frac{e^{-2x}}{-2} \right)_{x=y}^{\infty} (6 e^{-3y}) dy$$

$$= \int_0^{\infty} \frac{e^{-2y}}{2} \cdot 6 e^{-3y} dy = 3 \int_0^{\infty} e^{-5y} dy = \left(\frac{3}{5} \right)$$

#Q. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(y/x) = \begin{cases} \frac{1}{x - (-x)}, & -x \leq y \leq x \\ 0, & \text{otherwise} \end{cases}$$

Given $X = x$, the random variable Y is uniformly distributed on $[-x, x]$.

1) Find the joint PDF $f_{XY}(x, y)$

$$\text{i.e. } f(x, y) = \begin{cases} 1, & 0 \leq x \leq 1 \text{ \& } -x \leq y \leq x \\ 0, & \text{otherwise} \end{cases}$$

w.k. that, $P(A \cap B) = P(A/B) \cdot P(B)$

$$\Rightarrow f(x, y) = f(y/x) \cdot f(x) \\ = \frac{1}{2x} \cdot 2x = 1$$

$$f(x, y) = \begin{cases} 1, & |y| \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

[NAT]



2)

Find $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{x=|y|}^1 1 dx = 1 - |y|$

$$\text{So } f(y) = \begin{cases} 1 - |y|, & -1 \leq y \leq 1 \\ 0, & \text{else} \end{cases}$$

- #Q. A bank operates a drive-up and a walk-up window. Here
 X = the proportion of time the drive-up facility is in use
 Y = the proportion of time the walk-up window in use
 Say the manager has given the joint pdf based on his experience:

$$f(x, y) = \begin{cases} 6/5(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq \frac{1}{4} \\ 0 & \text{otherwise} \end{cases}$$

The probability that neither facility is busy more than one-quarter of the time is?

$$\begin{aligned}
 P\left(0 < x < \frac{1}{4}, 0 < y < \frac{1}{4}\right) &= \int_0^{1/4} \int_0^{1/4} f(x, y) dx dy = \int_0^{1/4} \int_0^{1/4} \frac{6}{5}(x + y^2) dx dy \\
 &= \frac{6}{5} \int_0^{1/4} \left(\frac{x^2}{2} + xy^2 \right) \Big|_{x=0}^{x=1/4} dy = \frac{6}{5} \int_0^{1/4} \left(\frac{1}{32} + \frac{y^2}{4} \right) dy = \frac{7}{640}
 \end{aligned}$$



THANK - YOU